

Knowledge and Language 2025/2026

literature Review

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I. ABSTRACT

This literature review explores the integration of Knowledge Graphs (KGs) and Retrieval-Augmented Generation (RAG) techniques for domain-specific question answering. The challenge we are trying to tackle is building a system that can reliably and accurately answer complex, domain-specific questions about Portuguese cuisine and other recipes using the food.com dataset [1]. We will use RecipeRAG [1], a KG-driven recipe generation framework, as reference for the methodologies to build a food chatbot. Definitions and concepts are supported by the class materials on semantic web framework (RDF, OWL, SPARQL), establishing the link between symbolic and knowledge graph representation.

II. CONNECTIONS WITH THE TOPICS COVERED BY THE COURSE

Large Language Models (LLMs) demonstrate strong generalization but fail in domain-specific reasoning due to three main limitations: (1) lack of contextual understanding, (2) incomplete factual knowledge, and (3) weak recall of precise facts [2]. This motivates the integration of symbolic knowledge (RDF/OWL-based graphs) with neural inference (LLMs and RAG) to ensure both reasoning and factual grounding. RDF can be serialized in formats such as N-TRIPLES, XML, JSON-LD, or Turtle, ensuring interoperability across systems [3]. Vocabularies define terms, relationships and ontologies define classes, hierarchies, and property restrictions. OWL (Web Ontology Language) can be used with RDF to express richer logical constraints, such as domain/range restrictions and class relations [3].

This symbolic foundation aligns with the three “knowledge-aware” dimensions of KG-augmented NLP methods outlined in [2]:

- KG-Augmented Retrieval: The use of structured queries (SPARQL) or textualized triples to retrieve relevant information for LLMs, comparable to RAG pipelines [4].
- KG-Augmented Reasoning and Controlled Generation: Incorporating graph reasoning steps (symbolic inference or logical validation) within the generation loop.
- Knowledge-Aware Training: Using KG-derived data to fine-tune LLMs, injecting factual consistency.
- Knowledge-Aware Validation: Employing SPARQL or rule-based checks to verify the factual correctness of generated answers.

This emphasizes the distinction between symbolic reasoning (FOL-based) and neural reasoning, emphasizing that symbolic models provide interpretability and factual precision, while neural models capture language variability and contextual generation.

III. RDFS/OWL KNOWLEDGE GRAPHS FOR REPRESENTING KNOWLEDGE FROM DATASETS

In RDF, knowledge is represented through triples — for example:

`<Bacalhau_a_Brás> <hasIngredient> <Egg>`

Each triple expresses a simple subject–predicate–object relationship, which can be enriched using RDFS with class and property hierarchies, domain and range constraints, and subclass relationships, allowing the representation of more structured and semantically rich knowledge. OWL further extends RDF/RDFS with expressive constructs for logical reasoning, such as cardinality restrictions, equivalence, disjointness, and complex class expressions, enabling more sophisticated inference over the data.

In [5], RecipeKG is constructed as a large-scale recipe knowledge graph that integrates ingredients, cooking techniques, and geographic regions, and enriches them with semantic annotations drawn from ontologies. By representing recipes, ingredients, and related concepts as URIs within a well-defined ontology, RecipeKG supports structured querying, semantic interoperability, and automated reasoning. For example, it can infer that all vegetarian recipes do not contain meat-based ingredients or suggest alternative ingredients based on class hierarchies. This structured representation allows applications such as intelligent recipe recommendation, dietary constraint checking, and multi-criteria search, which would be challenging to implement using unstructured data alone.

IV. KNOWLEDGE RETRIEVAL VIA NLP

Knowledge retrieval connects natural language input to structured or embedded KG representations. SPARQL is the declarative query language for RDF graphs [3], enabling semantic retrieval using triples and filters. SPARQL queries operate over RDF graphs, and their construction can be made in 3 different ways:

- The user, using the Chatbot, needs to input the SPARQL query directly or use a form-based interface that generates the query based on user selections.
- LLMs can be fine-tuned or prompted to generate SPARQL queries from natural language questions, using few-shot examples or templates to guide the generation.
- NLP tools like semantic parsers can convert natural language into SPARQL queries by identifying entities, intent, relations, restrictions and query structure.

Natural language interfaces require NLP-based query interpretation, extracting entities and intents (via tools like spaCy or BERT [6]) and mapping them to KG entities (URIs). In the system, BERT and spaCy are used together to understand user queries about recipes. BERT is fine-tuned to identify the user’s intent—such as finding a recipe, retrieving ingredients, or saving a recipe—which enables the system to correctly interpret what the user wants even when questions are phrased in different ways. BERT’s bidirectional transformer architecture provides deep contextual understanding, improving both the identification of user intents and entity disambiguation in complex queries. spaCy [7] is used to extract entities and linguistic features from user queries. Its efficient, rule and model-based pipeline enables fast and accurate identification of nouns, verbs, entities, and syntactic relations, helping map natural language input to knowledge graph entities and improving the system’s overall understanding of user intents. KGE-based retrieval [5] significantly outperforms text-based retrieval when handling multiple user constraints. As noted: “Performance declined as the number of criteria increased. . . Nevertheless, the results underscore the effectiveness of KGEs in supporting multi-criteria recipe retrieval, while simultaneously revealing the limitations of purely text-based embeddings.” Knowledge Graph Embeddings (KGEs) map entities and relations from symbolic graphs into continuous vector spaces, enabling reasoning and retrieval; **TransE** [8] represents relations as translation vectors such that the tail entity is approximated by the head plus the relation ($\mathbf{h} + \mathbf{r} \approx \mathbf{t}$) and works well for 1-to-1 relations but struggles with complex patterns, **RotatE** [9] models relations as rotations in the complex plane ($\mathbf{t} \approx \mathbf{h} \circ \mathbf{r}$) to naturally capture symmetry, antisymmetry, and inversion, and **QuatE** [10] extends this idea to quaternions ($\mathbf{t} \approx \mathbf{h} \otimes \mathbf{r}$) for 4D rotations, allowing modeling of more complex relational patterns.

V. RETRIEVAL-AUGMENTED GENERATION (RAG)

RAG, first formalized by [4], is a hybrid model that retrieves external documents as the input sequence x to retrieve text documents z and use them as additional context when generating the target sequence y . This approach separates retrieval from generation, improving factuality and reducing hallucinations. While the original RAG used text documents, later adaptations like the mentioned RecipeRAG [5] replace text retrieval with KG-based retrieval.

Lewis et al. (2020) [4] introduced the Retrieval-Augmented Generation (RAG) model, a hybrid architecture that combines parametric and non-parametric memory systems to improve

knowledge-intensive natural language generation. The parametric memory is represented by a pre-trained seq2seq model [11] (a transformer base model [12]), specifically BART, which stores implicit knowledge within its parameters. However, such knowledge is static, cannot be easily updated or interpreted, and may lead to factual inaccuracies or “hallucinations.” To address these issues, RAG incorporates a non-parametric memory — a dense vector index of Wikipedia built using the Dense Passage Retriever (DPR). Each passage is encoded as a dense vector, allowing the model to retrieve the top- k most relevant documents for a given query through Maximum Inner Product Search (MIPS):

$$p_\eta(z | x) \propto \exp(d(z)^\top q(x)) \quad (1)$$

where The retrieval component of RAG, denoted $p_\eta(z | x)$, represents the probability of retrieving a document z given a query x . Each document z is encoded into a dense vector representation $d(z)$, while the query x is encoded into a corresponding dense vector $q(x)$. The retriever computes the similarity between the query and each document using the **Maximum Inner Product Search (MIPS)** criterion, defined as

$$p_\eta(z | x) \propto \exp(d(z)^\top q(x)),$$

and retrieves the documents that maximize this inner product:

$$\arg \max_z d(z)^\top q(x).$$

This operation identifies the documents whose embeddings are most similar to the query embedding, ensuring that the most relevant passages are selected for the generator to use during response generation. During generation, RAG first uses DPR to retrieve relevant documents and then conditions the [13] generator on both the input and the retrieved passages. The final output is produced by marginalizing over all retrieved documents, effectively combining symbolic retrieval with neural generation.

Lewis et al. [4] proposed two variants of the Retrieval-Augmented Generation (RAG) model: *RAG-Sequence* and *RAG-Token*. *RAG-Sequence* conditions generation on a single retrieved document for the entire output, while *RAG-Token* allows token-level retrieval, enabling each generated token to draw information from different documents. For each token y_i , the model computes a weighted sum over the probabilities conditioned on each of the top- k retrieved documents, providing finer-grained factual grounding. This architecture allows RAG to generate more factual and interpretable text while maintaining the fluency of neural models. Unlike traditional language models that rely solely on fixed parametric memory, RAG can dynamically update its knowledge by replacing the document index, provide explicit evidence for its outputs through retrieved passages, and reduce hallucinations. Experimental results demonstrated *state-of-the-art performance* across knowledge-intensive tasks such as *Natural Questions*, *TriviaQA*, and *FEVER*, outperforming both parametric-only

models (e.g., T5, BART) and purely extractive retrieval systems (e.g., DPR, REALM). Moreover, RAG models produced more specific, diverse, and factually accurate generations, highlighting their effectiveness in combining retrieval-based reasoning with neural generation.

RecipeRAG [5] builds upon this foundation, substituting text retrieval with KG-based retrieval and applying it to domain-specific recipe generation. The authors state:

“We chose RotatE, the strongest-performing KGE model, to bootstrap the recipe generation process. The system retrieves the top-ranking recipes based on aggregated relevance scores, selects the top 5, and feeds them—together with user criteria—into the generator (DeepSeek-R1-0528).”

This pipeline combines the retrieval mechanism of [4] with structured KG enrichment, turning symbolic knowledge into context for generative models.

RAG thus operationalizes the Knowledge-Aware Inference paradigm [2], where external structured knowledge is used to enhance reasoning and factual correctness.

VI. CONCLUSION AND FUTURE WORK

The reviewed literature converges on the same insight: combining symbolic reasoning (RDF, OWL, SPARQL) and neural reasoning (KGE, RAG, LLMs) yields systems that are both factual and context-aware. Empirically, RecipeRAG [5] and RAG [4] demonstrate that KG-driven retrieval enhances generation quality, while the taxonomy of [2] clarifies how such systems reduce hallucination through Knowledge-Aware Inference, Learning, and Validation.

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